

Fast $(1, s)$ -Nucleus Decomposition via Static Clique Path Index

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Submission #XXX

Abstract

Exact $(1, s)$ -nucleus decomposition ranks vertices by triangle, 4-clique, and larger clique witnesses, but high- s decomposition is expensive because the number of witnesses can be enormous. The Clique Path Index compactly encodes these witnesses, but the general (r, s) framework maintains the index as mutable residual state during peeling. We show that this mutability is unnecessary for the vertex-centric case $r = 1$: vertex peeling removes vertices but does not delete edges, so every CPI path remains structurally valid. This invariant turns exact decomposition into counter maintenance over a static CPI, which yields SPIN, a direct static-path iteration, SPIN*, its event-driven optimization, and a parallel builder designed for the shifted construction bottleneck. On 505 matched successful configurations, the static pipeline achieves a $6.16\times$ geometric-mean speedup, a $39.41\times$ maximum speedup, and lower peak RSS than the mutable baseline. The result shows that specializing the state model for $(1, s)$ can make exact high-order vertex-centric decomposition practical without enumerating all s -cliques.

CCS Concepts

• **Theory of computation** → **Graph algorithms analysis**; • **Information systems** → **Data mining**.

Keywords

nucleus decomposition, clique counting, cohesive subgraphs, hierarchy

ACM Reference Format:

Anonymous Authors. 2027. Fast $(1, s)$ -Nucleus Decomposition via Static Clique Path Index. In *Proceedings of the 2027 ACM SIGMOD International Conference on Management of Data (SIGMOD '27)*. ACM, New York, NY, USA, 15 pages. <https://doi.org/XXXXXXX.XXXXXXX>

1 Introduction

Core decomposition ranks vertices by the amount of cohesive structure that still surrounds them after weak vertices are removed [1, 12, 14]. The classical k -core uses edges as the witness of cohesion. Many graph mining tasks need a stronger witness, because a group may pass an edge-degree threshold while containing few triangles, 4-cliques, or larger coordinated patterns. The $(1, s)$ -nucleus decomposition [11] gives this vertex-centric higher-order ranking. For a fixed s , it assigns each vertex the largest value k for which the vertex belongs to a maximal region where every vertex participates in at least k internal s -cliques. At $s = 2$, this is the k -core. At

$s = 3$, it ranks vertices by triangle-supported cohesion. At larger s , it exposes progressively tighter clique-based kernels.

Exact high- s decomposition is hard because the number of s -cliques can be enormous. The current state of the art avoids explicit enumeration with the Clique Path Index, or CPI [9]. A CPI path stores a hold set and a pivot set. It compactly represents all s -cliques formed by taking every hold vertex and choosing the remaining vertices from the pivot set. This representation made general (r, s) -nucleus decomposition practical, but the general algorithm treats the index as mutable residual state during peeling. It stores a mutable recursion tree, per-path vertex sets, reverse maps, and a priority queue. This generality is costly. It is also necessary for the general case, because when $r \geq 2$, peeling can delete edges or higher-order objects and can break the clique structure that a path stores.

The vertex-centric case has a stronger invariant. When $r = 1$, peeling removes vertices but does not delete edges. Therefore, a set of vertices that formed a clique when the CPI was built remains a clique throughout the peel. The path may stop contributing because a required hold vertex has been removed. Or its contribution may shrink because one pivot vertex has been removed. But the stored hold and pivot sets do not need to be rewritten. The dynamic state of a path is only whether it can still contribute, how many pivots each encoded clique must choose, and how many pivot vertices have not yet been removed.

This observation is the thesis of the paper: for $(1, s)$ -nucleus decomposition, vertex peeling preserves the CPI structure, so exact decomposition can be performed over a static clique-path index using only counter updates. Theorem 3.1 states the static path-event rule in counter form. Lemma 3.1 turns each peel event into a closed-form binomial support delta. Together they replace mutable residual-index maintenance with arithmetic over a static incidence layout.

Once the index is static, the core computation becomes a fixed point over threshold support. The numerical operator is the same one used in core fixed-point characterizations [5, 7, 8], but its items are clique witnesses rather than neighbor vertices. We call the direct solver SPIN, for *Static Path Index for Nucleus decomposition*. SPIN repeatedly recomputes every vertex value from static CPI witness counts until the values stop changing. This direct form is useful because it states the computation without residual-index mutation, but it still scans the whole graph after each pass.

SPIN* is the event-driven optimization of SPIN. It computes the same fixed point, but changes the schedule. Instead of full-graph passes, it finalizes vertices in nondecreasing value order and revisits only vertices whose path counters changed. Thus the star marks an execution improvement, not a different decomposition objective.

Making peeling cheap changes the bottleneck. After support maintenance no longer mutates the residual CPI, construction becomes a dominant part of end-to-end time on many benchmark instances. This motivates ParaBuildCPI as part of the same chain rather than as a separate optimization. The degeneracy-ordered

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SIGMOD '27, June 2027, TBD

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ACM ISBN 978-1-XXXX-XXXX-X/27/06...\$15.00
<https://doi.org/XXXXXXX.XXXXXXX>

seed loop already gives disjoint ownership of s -cliques. Because the downstream consumer needs only a static path list and incidence arrays, each thread can stream emitted paths into a local arena and merge the arenas deterministically.

The same deletion order also gives a core-value hierarchy. SPIN* emits vertices in nondecreasing κ_s , because its bucket cursor never retreats. Let $V_k = \{v : \kappa_s(v) \geq k\}$. The edge-connected components of $G[V_k]$ can therefore be recovered by scanning this deletion log in reverse. BuildHier uses only the input graph and union-find to extract this elder-rule join tree.

The contributions are a dependency chain.

- **Static-CPI state for $(1, s)$.** We derive a static-counter rule for vertex peeling: exact peeling never rewrites stored hold and pivot sets, and each event becomes a path death or pivot-counter decrement with a closed-form support delta (Theorem 3.1, Lemma 3.1).
- **SPIN and SPIN* over the static index.** We formulate SPIN as the direct threshold-support iteration on CPI witness counts, and SPIN* as the event-driven optimization that computes the same fixed point without repeated global recomputations (Algorithms 2 and 3, Lemma 4.3).
- **Bottleneck-shifted static construction.** Once peeling becomes counter-only, CPI construction becomes the dominant remaining cost on many instances; ParaBuildCPI exploits the static output contract by streaming paths into thread-local arenas and merging them deterministically without hot-path synchronization (Algorithm 4).
- **Core-value hierarchy from the deletion log.** We show that the sorted log produced by SPIN* is enough to recover the edge-connected super-level join tree in $O((n + m)\alpha(n))$ time using only the input graph (Algorithm 5).
- **Experimental and application evidence.** We compare against the mutable-CPI baseline, report runtime and memory on 762 matched benchmark cells, measure phase behavior and parallel construction, and use case studies to evaluate whether the $r = 1$ specialization is practically useful.

The rest of the paper follows this chain. Section 2 defines the problem, the CPI, and the mutable baseline. Section 3 derives the static-CPI invariant from the vertex-removal rule. Section 4 derives SPIN and SPIN* from the same threshold-support fixed point. Section 5 parallelizes construction after peeling becomes cheap. Section 6 extracts the hierarchy from the deletion log. Sections 7 and 8 report performance and application evidence. Sections 9 and 10 position and conclude the paper.

2 Preliminaries

The introduction stated the paper's central invariant in informal terms. To prove and use it, we need precise objects: vertex support, $(1, s)$ -nucleus core values, CPI paths, and the mutable baseline that the paper specializes. This section fixes that vocabulary and ends at the point where the $r = 1$ simplification can be stated cleanly.

Graph notation. We work with a simple undirected graph $G = (V, E)$. For a vertex v , $N_G(v) = \{u \in V \mid (u, v) \in E\}$ is its neighbor set and $\deg_G(v) = |N_G(v)|$ is its degree. For $X \subseteq V$, $G[X]$ is the subgraph induced by X . The implementation stores G in CSR form with an offsets array and an edges array.

Definition 2.1 (k -Clique). A k -clique is a vertex set $C \subseteq V$ with $|C| = k$ such that every two distinct vertices in C are adjacent.

Definition 2.2 (s -Clique Support). Given $s \geq 2$, the s -clique support of a vertex v in G is

$$\deg^{(s)}(v) = |\{S \subseteq V \mid v \in S, |S| = s, S \text{ is a clique in } G\}|.$$

Support is always evaluated inside the current induced subgraph when a peel level is being defined. When the graph is clear, we write $\deg^{(s)}(v)$ without a subscript.

Definition 2.3 (k -Admissible s -Clique). Given $s \geq 2$ and $k \geq 1$, an s -clique S in G is k -admissible if every vertex $v \in S$ satisfies $\deg^{(s)}(v) \geq k$ in the subgraph under consideration.

Definition 2.4 (k -(1, s)-Nucleus). Let $\mathcal{S}_k$ be the set of k -admissible s -cliques of G . For any $\mathcal{S} \subseteq \mathcal{S}_k$, let $G[\mathcal{S}] = G[\bigcup_{S \in \mathcal{S}} S]$. A subgraph $G_k = G[\mathcal{S}]$ is a k -(1, s)-nucleus if:

- (1) G_k is s -connected: for any two vertices $u, v \in V(G_k)$, there is a sequence $u = v_1, \dots, v_t = v$ such that each pair $\{v_i, v_{i+1}\}$ is contained in some $S_i \in \mathcal{S}$;
- (2) G_k is maximal: no proper superset $\mathcal{S}' \supset \mathcal{S}$ still satisfies the first condition.

Thus a k -(1, s)-nucleus is a maximal region in which every surviving vertex has at least k internal s -clique witnesses and the region is connected through those witnesses.

Definition 2.5 (Core Number). Given $s \geq 2$, the $(1, s)$ -nucleus core number of a vertex v , denoted $\kappa_s(v)$, is the largest k for which v belongs to some k -(1, s)-nucleus of G .

When $s = 2$, κ_s is the classical core number. When $s = 3$, it ranks vertices by triangle-supported cohesion.

The Clique Path Index. The Clique Path Index, or CPI, compactly represents s -cliques without listing them one by one [9].

Definition 2.6 (CPI Path). A CPI path is a pair $\mathcal{P} = (V_h(\mathcal{P}), V_p(\mathcal{P}))$ of disjoint vertex sets such that $V_h(\mathcal{P}) \cup V_p(\mathcal{P})$ is a clique in G and $|V_h(\mathcal{P})| \leq s \leq |V_h(\mathcal{P})| + |V_p(\mathcal{P})|$. The set $V_h(\mathcal{P})$ is the *hold set*: every encoded s -clique contains all of these vertices. The set $V_p(\mathcal{P})$ is the *pivot set*: an encoded s -clique chooses vertices from this pool. The *pivot requirement* of $\mathcal{P}$ is $\eta_{\mathcal{P}} = s - |V_h(\mathcal{P})|$, the number of pivot vertices needed to complete the hold set into an s -clique. The set of s -cliques encoded by $\mathcal{P}$ is

$$C(\mathcal{P}) = \{V_h(\mathcal{P}) \cup P' \mid P' \subseteq V_p(\mathcal{P}), |P'| = \eta_{\mathcal{P}}\}.$$

Its cardinality is $\binom{|V_p(\mathcal{P})|}{\eta_{\mathcal{P}}}$.

The CPI is a set $\mathcal{P}$ of paths such that every s -clique is encoded by exactly one path. We write

$$\Sigma = \sum_{\mathcal{P} \in \mathcal{P}} (|V_h(\mathcal{P})| + |V_p(\mathcal{P})|)$$

for the vertex-path incidence count. We also write $\deg_{\Sigma}(v) = |\{\mathcal{P} \mid v \in V_h(\mathcal{P}) \cup V_p(\mathcal{P})\}|$, $d_{\max} = \max_v \deg_{\Sigma}(v)$, and $\kappa_{\max} = \max_v \deg^{(s)}(v)$. The CPI is built by a degeneracy-ordered Bron-Kerbosch search with Tomita pivoting [2, 4, 9, 13].

Lemma 2.1 (Support Formula [9]). For every $v \in V$,

$$\deg^{(s)}(v) = \sum_{\mathcal{P} : v \in V_h(\mathcal{P})} \binom{|V_p(\mathcal{P})|}{\eta_{\mathcal{P}}} + \sum_{\mathcal{P} : v \in V_p(\mathcal{P})} \binom{|V_p(\mathcal{P})| - 1}{\eta_{\mathcal{P}} - 1}.$$

The mutable peeling baseline. The standard peeling loop repeatedly removes a vertex of minimum residual support, assigns its core value, and propagates support losses to the remaining vertices [1, 9]. For general (r, s) -nucleus decomposition, the baseline keeps the CPI as mutable residual state. Its implementation uses a mutable Bron–Kerbosch recursion tree, per-path vertex hash sets, reverse vertex-to-path maps, and a heap. This state is needed in the general setting because removing an r -clique with $r \geq 2$ may delete edges or higher-order objects that invalidate stored path structure. In that baseline, each peel event can pay for hash erases, tree deletion or shrinking, and heap decrease-key operations. The next section shows that the $r = 1$ case does not need this mutable state.

Problem statement. Given G and $s \geq 2$, compute $\kappa_s(v)$ for every $v \in V$. Section 6 treats hierarchy construction after the core values have been computed.

3 Static CPI for $r = 1$

The previous section ended with the mutable residual-index baseline. The question now is whether that residual state is necessary when the peeled object is a vertex. For $r = 1$, the answer is no. Vertex peeling removes vertices from the active subgraph, but it does not delete edges among the vertices that remain. Therefore, the stored clique $V_h(\mathcal{P}) \cup V_p(\mathcal{P})$ of every path remains structurally valid for as long as its active vertices are considered. Only the contribution of the path changes.

Static structure and dynamic counters. A path stores a fixed recipe for its encoded s -cliques. The hold set $V_h(\mathcal{P})$ is the part that every encoded clique must contain. The pivot set $V_p(\mathcal{P})$ is the pool from which the path chooses the remaining vertices. The pivot requirement $\eta_{\mathcal{P}} = s - |V_h(\mathcal{P})|$ says how many still-effective pivots must be chosen from that pool. These three objects are static. During peeling, the algorithm does not edit the stored hold or pivot sets. The dynamic state is only a liveness bit and an integer $p_{\mathcal{P}}$, the number of pivots in $V_p(\mathcal{P})$ that have not yet been peeled. If a peeled vertex is a hold, every encoded s -clique contains that vertex, so the path contributes nothing after the event. If a peeled vertex is a pivot, the path remains the same symbolic object, but $p_{\mathcal{P}}$ decreases by one. This is the whole state transition. The next theorem gives the counter form we need: a peel event either kills a path contribution or reduces an effective pivot count.

Theorem 3.1 (Vertex Removal on CPI, Counter Form). *Let $\mathcal{P}$ be a path of the CPI of G , and let $v \in V$ be removed from G .*

- (1) *If $v \in V_h(\mathcal{P})$, $\mathcal{P}$ encodes no surviving s -clique.*
- (2) *If $v \in V_p(\mathcal{P})$, the surviving s -cliques are encoded by the same $V_h(\mathcal{P})$ together with $V_p(\mathcal{P}) \setminus \{v\}$. Their count depends only on $|V_p(\mathcal{P})|$. A counter $p_{\mathcal{P}} = |V_p(\mathcal{P})|$ decremented by one is therefore support-equivalent to deleting v from $V_p(\mathcal{P})$. If $p_{\mathcal{P}} - 1 < \eta_{\mathcal{P}}$, no surviving s -clique is encoded.*
- (3) *Otherwise, $\mathcal{P}$ is unaffected.*

The dynamic state of $\mathcal{P}$ during peeling reduces to an integer counter $p_{\mathcal{P}}$, the fixed pivot requirement $\eta_{\mathcal{P}}$, and a liveness bit. The stored sets $V_h(\mathcal{P})$ and $V_p(\mathcal{P})$ are never modified.

PROOF. $\mathcal{P}$ encodes $C(\mathcal{P}) = \{V_h(\mathcal{P}) \cup P' \mid P' \subseteq V_p(\mathcal{P}), |P'| = \eta_{\mathcal{P}}\}$. If $v \in V_h(\mathcal{P})$, every encoded s -clique contains v , so no encoded clique survives after v is removed. If $v \in V_p(\mathcal{P})$, the surviving

encoded cliques are exactly

$$\{V_h(\mathcal{P}) \cup P' \mid P' \subseteq V_p(\mathcal{P}) \setminus \{v\}, |P'| = \eta_{\mathcal{P}}\}.$$

This family has cardinality $\binom{|V_p(\mathcal{P})| - 1}{\eta_{\mathcal{P}}}$. By Lemma 2.1, every support contribution of the path depends on the pivot set only through this cardinality and the corresponding pivot cardinalities. Thus replacing the physical deletion from $V_p(\mathcal{P})$ by the counter update $p_{\mathcal{P}} \leftarrow p_{\mathcal{P}} - 1$ gives the same support arithmetic. The family is empty exactly when fewer than $\eta_{\mathcal{P}}$ effective pivots remain. If $v \notin V_h(\mathcal{P}) \cup V_p(\mathcal{P})$, no encoded clique on $\mathcal{P}$ contains v , so the encoded family is unchanged. $\square$ $\square$

Boundary of the invariant. The theorem is specific to $r = 1$. When $r \geq 2$, a peel event removes an edge or a higher-order clique from the residual object. That event can break the clique represented by $V_h(\mathcal{P}) \cup V_p(\mathcal{P})$. In that setting, a path may need to be split, shrunk, or rebuilt. The mutable CPI machinery of [9] remains the appropriate general tool, and this paper does not subsume it.

Closed-form support delta. Theorem 3.1 leaves only cardinality changes. The support lost by any other live vertex on the path is therefore a binomial difference.

Lemma 3.1 (Closed-form Support Delta). *Let $\mathcal{P}$ be a live path with $p \geq \eta$ before an event that removes $d \geq 1$ live pivots from $\mathcal{P}$. For every other alive vertex u incident to $\mathcal{P}$:*

$$\begin{aligned} \Delta_{\text{hold}}(p, \eta, d) &= \binom{p}{\eta} - \binom{p-d}{\eta} & \text{if } u \in V_h(\mathcal{P}), \\ \Delta_{\text{pivot}}(p, \eta, d) &= \binom{p-1}{\eta-1} - \binom{p-d-1}{\eta-1} & \text{if } u \in V_p(\mathcal{P}). \end{aligned}$$

A path-killing event, caused by a hold peel or by $p - d < \eta$, is the limiting case in which the entire contribution vanishes: $\binom{p}{\eta}$ for surviving holds and $\binom{p-1}{\eta-1}$ for surviving pivots.

PROOF. Apply Lemma 2.1 before and after the event. A hold u contributes $\binom{p}{\eta}$ before the event and $\binom{p-d}{\eta}$ after the event. A pivot u occupies one pivot slot. It contributes $\binom{p-1}{\eta-1}$ before the event and $\binom{p-d-1}{\eta-1}$ after the event. Subtracting the after value from the before value gives the two formulas. $\square$ $\square$

Running example. Figure 1 shows the running example at $s = 3$: two K_4 blocks share edge (v_3, v_4) , and two pendant triangles attach at v_2 and v_6 . The six CPI paths in Table 1 encode the ten triangle witnesses through stored V_h, V_p sets and the binomial count $\binom{|V_p|}{s - |V_h|}$. The $(1, 3)$ -core values fall into two bands: $\kappa_3(v_i) = 3$ for $i \in \{1, \dots, 6\}$ (the two K_4 blocks) and $\kappa_3(v_i) = 1$ for $i \in \{7, \dots, 10\}$ (the pendant triangles); this is the partition shown by the node shading in Figure 1. The example exercises both events of Theorem 3.1: a path dies when a hold vertex is removed, and a path's pivot counter shrinks when a pivot vertex is removed.

Example 3.1 (Peel Event Walk-through). Consider the path $\mathcal{P}_3 = (\{v_1\}, \{v_2, v_3, v_4\})$ in Table 1, with $p = 3$ and $\eta = 2$. Peeling v_2 as a pivot triggers Lemma 3.1 with $d = 1$. The surviving hold v_1 loses $\binom{3}{2} - \binom{2}{2} = 2$ from its support. The surviving pivots v_3 and v_4 each lose $\binom{2}{1} - \binom{1}{1} = 1$. The stored set $V_p(\mathcal{P}_3)$ is not modified. Only $p_{\mathcal{P}_3}$ changes from 3 to 2.

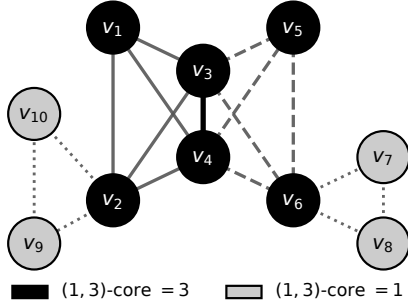


Figure 1: Running example at $s = 3$: two K_4 blocks share edge (v_3, v_4) (heavy black); pendant triangles attach at v_2 and v_6 (dotted edges). Black nodes have $(1, 3)$ -core value 3; light-gray nodes have value 1. Edge styles distinguish the two K_4 blocks (solid vs. dashed) from the pendant triangles (dotted).

Table 1: The six CPI paths for the running example at $s = 3$. Total binomial count $\sum \binom{|V_P|}{\eta} = 10$ matches the ten triangles in the graph.

Path	V_h	V_p	$\eta = s - V_h $	$\binom{ V_P }{\eta}$
$\mathcal{P}_1$	$\{v_7\}$	$\{v_6, v_8\}$	2	1
$\mathcal{P}_2$	$\{v_9\}$	$\{v_2, v_{10}\}$	2	1
$\mathcal{P}_3$	$\{v_1\}$	$\{v_2, v_3, v_4\}$	2	3
$\mathcal{P}_4$	$\{v_2\}$	$\{v_3, v_4\}$	2	1
$\mathcal{P}_5$	$\{v_3\}$	$\{v_4, v_5, v_6\}$	2	3
$\mathcal{P}_6$	$\{v_4\}$	$\{v_5, v_6\}$	2	1
Total				10

4 Computing Core Values over Static CPI

Static CPI leaves a support-counting problem instead of an index-maintenance problem. Section 3 showed that the threshold witnesses of a vertex can be counted from unchanged CPI paths and path counters. The remaining question is how to turn those support values into final core numbers. This section turns the core condition into a vertex update, then derives two solvers from the same update. SPIN applies the update by repeated full passes over the vertices. SPIN* computes the same result in peel order and revisits only vertices whose path counters changed.

4.1 Vertex updates from static witnesses

The static-CPI invariant tells us how to count witnesses without rebuilding residual paths. To compute core values, we also need a rule for lowering a vertex value from the witnesses that remain. The rule asks for the largest support level that the queried vertex can still justify. At level k , the vertex needs at least k encoded clique witnesses whose other vertices can also sustain level k . For a tentative value vector, define the one-vertex update

$$\Phi_h(v) = \max\{k \geq 0 \mid W_v(h, k) \geq k\}.$$

Here $W_v(h, k)$ is the number of encoded s -clique witnesses through v whose other vertices have value at least k . For clique size two, the witnesses are edges, so the update becomes the standard core fixed-point test over neighbors [5, 7, 8]. For larger clique sizes, the counted items are clique witnesses, not distinct neighbor vertices. The next

Algorithm 1: VertexSupport(v, h) — one-vertex update from static clique witnesses.

Input: Vertex $v \in V$; current h -value vector h ; CPI $\mathcal{P}$.
Output: The threshold-support value $\Phi_h(v)$.

```

1  $C[0, \dots, h[v]] \leftarrow 0$ ; // threshold witness counts
2 foreach path  $\mathcal{P} \in \mathcal{P}$  through  $v$  do
3   for  $k \leftarrow 1$  to  $h[v]$  do
4      $C[k] \leftarrow C[k] + W_v(\mathcal{P}, h, k)$ ; // Lemma 4.1
5 for  $k \leftarrow h[v]$  to 1 do
6   if  $C[k] \geq k$  then return  $k$ ;
7 return 0
```

lemma counts $W_v(h, k)$ from CPI paths without enumerating those witnesses.

Lemma 4.1 (Per-path Witness Count). *Fix a vertex v , a current vector h , and a threshold $k \geq 1$. For a path $\mathcal{P}$ with $v \in V_h(\mathcal{P}) \cup V_p(\mathcal{P})$, let $H_k(\mathcal{P})$ and $Q_k(\mathcal{P})$ be the number of vertices in $V_h(\mathcal{P}) \setminus \{v\}$ and $V_p(\mathcal{P}) \setminus \{v\}$, respectively, whose current h -value is at least k . The number $W_v(\mathcal{P}, h, k)$ of encoded s -cliques through v in which every other vertex has h -value at least k is:*

- (1) if $H_k(\mathcal{P}) < |V_h(\mathcal{P}) \setminus \{v\}|$, then $W_v(\mathcal{P}, h, k) = 0$;
- (2) if $v \in V_h(\mathcal{P})$, then $W_v(\mathcal{P}, h, k) = \binom{Q_k(\mathcal{P})}{\eta_P}$;
- (3) if $v \in V_p(\mathcal{P})$, then $W_v(\mathcal{P}, h, k) = \binom{Q_k(\mathcal{P})}{\eta_P - 1}$.

The total threshold count is $W_v(h, k) = \sum_{\mathcal{P} \ni v} W_v(\mathcal{P}, h, k)$. $\Phi_h(v)$ is the largest k such that $W_v(h, k) \geq k$.

PROOF. Each encoded s -clique through v has the form $V_h(\mathcal{P}) \cup P'$, where $P' \subseteq V_p(\mathcal{P})$ and $|P'| = \eta_P$. If any other hold vertex fails the threshold k , no encoded clique through v can qualify. If v is a hold, all η_P pivot slots must be filled by threshold-passing pivots. If v is a pivot, v occupies one pivot slot and the remaining $\eta_P - 1$ slots must be filled by threshold-passing pivots. Summing the qualifying counts over paths gives $W_v(h, k)$. $\square$ $\square$

VertexSupport(v, h) computes this update for one vertex. During one call, the algorithm builds only local threshold state for v . After any prefix of the paths through v , that local state contains exactly the witnesses contributed by the scanned paths. Scanning the next path adds the binomial counts from Lemma 4.1, and the largest threshold with enough witnesses is the updated value. Because the call reads the static CPI but never mutates a path, the same witness rule can be used by both solvers below.

A direct implementation does not enumerate clique witnesses. It evaluates the same threshold counts from the values stored on each scanned path, using sorting or buckets. In the stated implementation, one call costs $O(\deg_\Sigma(v) \log \deg_\Sigma(v))$. The call reads the static CPI and the vector h , and it mutates no path.

4.2 Direct h -Index Iteration

VertexSupport defines the threshold-support equation for one vertex. SPIN is the most literal solver: it scans all vertices, computes a next value for each vertex from the previous value vector, and repeats until a full scan makes no change. Its purpose is to expose the fixed-point computation directly. The only mutable data are the current value vector and the next value vector. The current vector starts at the initial s -clique support, so every entry is an upper bound on the final core value. Each pass can only lower entries. A fixed point means every vertex value is consistent with

Algorithm 2: SPIN — direct static-path fixed-point iteration.

Input: Graph G , clique size s , CPI $\mathcal{P}$.
Output: $\kappa_s(v)$ for every $v \in V$.

```

1 foreach  $v \in V$  do
2    $h[v] \leftarrow \deg^{(s)}(v)$ ;           // closed form from Lemma 2.1
3 changed  $\leftarrow$  true;
4 while changed do
5   changed  $\leftarrow$  false;  $h_{\text{new}} \leftarrow h$ ;
6   foreach  $v \in V$  in a fixed order do
7      $h_{\text{new}}[v] \leftarrow \text{VertexSupport}(v, h)$ ;
8     if  $h_{\text{new}}[v] \neq h[v]$  then changed  $\leftarrow$  true;
9    $h \leftarrow h_{\text{new}}$ ;
10 foreach  $v \in V$  do
11    $\kappa_s(v) \leftarrow h[v]$ ;
12 return  $\kappa_s$ 

```

the witnesses whose other vertices have high enough current value. This makes SPIN the reference formulation for the static-CPI computation, even though repeated full scans can be expensive.

Algorithm 2 keeps the implementation close to the equations. During one pass, every call to `VertexSupport` reads the old vector and writes into the next vector. The swap at the end of the pass avoids order dependence. A pass with no changed entry is the certificate that the fixed point has been reached.

Example 4.1 (h -Index Refresh on the Running Example). Initialise $h[v] \leftarrow \deg^{(s)}(v)$ using Lemma 2.1; on the running example $h[v_4] = 7$ (contributions from $\mathcal{P}_3, \mathcal{P}_4$ as pivot, $\mathcal{P}_5, \mathcal{P}_6$ as hold; see Table 1). The first call `VertexSupport`(v_4, h) scans the four paths through v_4 and counts threshold witnesses via Lemma 4.1. At threshold $k = 3$, paths $\mathcal{P}_3$ and $\mathcal{P}_5$ together contribute $W_{v_4}(h, 3) = 3$ witnesses with all other path vertices at $h \geq 3$, so the largest k with $W_{v_4}(h, k) \geq k$ is $\Phi_h(v_4) = 3$, the final $\kappa_3(v_4)$. SPIN reaches the fixed point in a small number of synchronous passes, but each pass scans every vertex regardless of whether its value can still change — the redundancy that SPIN* eliminates in Section 4.3.

The proof below connects the fixed point to $(1, s)$ -nucleus core values. It is the reason SPIN can serve as a correctness reference for the event-driven algorithm.

Lemma 4.2 (Fixed Point Correctness). *Initialised at $h^{(0)}[v] = \deg^{(s)}(v)$, Algorithm 2 converges in at most $1 + n\kappa_{\max}$ passes to $h[v] = \kappa_s(v)$ for every $v \in V$. The sequence $h^{(t)}[v]$ is monotonically non-increasing.*

PROOF. At any pass, $h^{(t+1)}[v] \geq k$ exactly when at least k incident s -cliques have all other vertices with $h^{(t)} \geq k$. The initial value $\deg^{(s)}(v)$ is the largest possible support value for v , so the update cannot increase $h[v]$. The values are non-negative integers bounded by $\kappa_{\max}$, so the sequence converges. At a fixed point h^* , the set $\{v \mid h^*[v] \geq k\}$ is exactly the maximal vertex set in which each vertex has at least k incident s -cliques inside the set. These sets define the $(1, s)$ -nucleus levels. Therefore $h^*[v] = \kappa_s(v)$. The crude pass bound follows because across all vertices the sum of integer values can decrease at most $n\kappa_{\max}$ times before the final no-change pass. $\square$

Theorem 4.1 (SPIN Complexity). *Algorithm 2 computes κ_s in $O(T \cdot \Sigma \cdot \log d_{\max})$ work, where $T \leq 1 + n\kappa_{\max}$ is the number of passes.*

Algorithm 3: SPIN* — core values via single-pass bucket-queue peeling.

Input: Graph G , clique size s , CPI $\mathcal{P}$.
Output: $\kappa_s(v)$ for every $v \in V$.

```

1 foreach path  $\mathcal{P} \in \mathcal{P}$  do
2    $p_{\mathcal{P}} \leftarrow |V_{\mathcal{P}}(\mathcal{P})|$ ;  $\eta_{\mathcal{P}} \leftarrow s - |V_h(\mathcal{P})|$ ;  $\text{alive}_{\mathcal{P}} \leftarrow (p_{\mathcal{P}} \geq \eta_{\mathcal{P}})$ ;
3 foreach  $v \in V$  do
4    $h[v] \leftarrow \deg^{(s)}(v)$ ; insert  $v$  into bucket  $B[h[v]]$ ;
5  $k \leftarrow 0$ ;  $V_{\text{fin}} \leftarrow \emptyset$ ;
6 while  $|V_{\text{fin}}| < n$  do
7   advance  $k$  until  $B[k] \neq \emptyset$ ; pop  $v$  from  $B[k]$ ;
8    $\kappa_s(v) \leftarrow k$ ; append  $(v, k)$  to the deletion log;  $V_{\text{fin}} \leftarrow V_{\text{fin}} \cup \{v\}$ ;
9    $\text{Aff} \leftarrow \emptyset$ ;
10  foreach  $\mathcal{P}$  with  $\text{alive}_{\mathcal{P}}$  and  $v \in V_h(\mathcal{P}) \cup V_p(\mathcal{P})$  do
11     $\text{Aff} \leftarrow \text{Aff} \cup ((V_h(\mathcal{P}) \cup V_p(\mathcal{P})) \setminus \{v\})$ ;
12    if  $v \in V_h(\mathcal{P})$  then
13       $\text{alive}_{\mathcal{P}} \leftarrow$  false;           // path death
14    else
15       $p_{\mathcal{P}} \leftarrow p_{\mathcal{P}} - 1$ ;
16      if  $p_{\mathcal{P}} < \eta_{\mathcal{P}}$  then  $\text{alive}_{\mathcal{P}} \leftarrow$  false;
17  foreach  $u \in \text{Aff} \setminus V_{\text{fin}}$  do
18     $y \leftarrow \max\{k, \text{VertexSupport}(u, h)\}$ ;           // implemented
19    incrementally with Lemma 3.1
20    if  $y < h[u]$  then move  $u$  from  $B[h[u]]$  to  $B[y]$ ;  $h[u] \leftarrow y$ ;
21 return  $\kappa_s$ 

```

PROOF. Each pass calls `VertexSupport`(v, h) for every vertex v . The cost of one call is $O(\deg_{\Sigma}(v) \log \deg_{\Sigma}(v))$. Summing over vertices gives $O(\Sigma \log d_{\max})$ work per pass. Lemma 4.2 bounds the number of passes. $\square$

4.3 Eliminating Redundant Sweeps

SPIN is useful because it exposes the equations, but its full passes recompute many vertices whose values can no longer change. Once a vertex reaches its final peel level, later events cannot increase it. SPIN* removes this redundancy by using the same nondecreasing peel order as classical core decomposition. The algorithm keeps the same value semantics as SPIN, but it updates them with local path events instead of global passes. For each path, it stores whether the path is still live, how many pivot vertices have not been finalized, and the fixed pivot requirement. It also keeps a finalized set and a bucket queue keyed by the current value. When one minimum vertex is popped, only paths containing that vertex can change. Theorem 3.1 says how each such path changes, and Lemma 3.1 gives the exact support loss for the unfinalized path-mates. Thus the invariant is local: unfinalized vertex values reflect the current live path-counter contributions, clipped below by the current peel level.

SPIN* is an optimization of SPIN, not a different decomposition objective. It keeps the same witness equation but changes the schedule. Instead of scanning every vertex after every pass, it updates only the vertices whose path contributions changed. Algorithm 3 outputs the core values and emits a sorted deletion log as a byproduct. Its running time depends on U , the number of incremental support-update events that actually occur, rather than on the number of full scans.

The pseudocode writes the affected update as `VertexSupport` to keep the fixed-point semantics visible. The implementation evaluates the changed path contribution by the closed-form delta from Lemma 3.1. The total number of such evaluations is U . Buckets

maintain the minimum- h vertex because values only decrease and are bounded by $\kappa_{\max}$ [1].

Lemma 4.3 (SPIN* Equivalence). *Algorithm 3 returns the same κ_s values as Algorithm 2.*

PROOF. The algorithms use the same witness equations. SPIN evaluates them by repeated full passes over the vertex set. SPIN* evaluates them in the standard nondecreasing peel order. The loop invariant for SPIN* is that every unfinalized vertex u has the same value it would obtain from the current live witness multiset under the fixed-point equations, except that values below the current peel level k are clipped to k . The invariant is true initially because $h[u] = \deg^{(s)}(u)$. When v is finalized, only paths containing v can change any remaining witness multiset. Theorem 3.1 gives the exact state transition for each such path. Lemma 3.1 gives the exact support loss for every affected path-mate. Thus the updates to Aff restore the invariant, while vertices outside Aff need no update. Popping a minimum bucket finalizes the next vertex whose fixed-point value cannot exceed the current level. By Lemma 4.2, that fixed point is κ_s . $\square$

Theorem 4.2 (SPIN* Complexity). *Algorithm 3 runs in $O(\Sigma + U + \kappa_{\max} + n)$ time, where U is the total number of incremental support-refresh events over all affected path-mates.*

PROOF. Initialization of vertex values and path counters costs $O(n + \Sigma)$. Each vertex-path incidence is visited when its vertex is finalized, so path-event processing costs $O(\Sigma)$. Each affected incremental refresh contributes one unit to U . Bucket moves and bucket pops are $O(1)$ amortized. The bucket cursor advances at most $\kappa_{\max}$ times. The total time is therefore $O(\Sigma + U + \kappa_{\max} + n)$. $\square$

The mutable baseline pays heap and residual-index maintenance costs on the same logical peel events. The experiments in Section 7 measure the impact of replacing those operations with static path counters.

5 Parallel CPI Construction

Section 4 reduced residual support maintenance to static path-counter updates. Once peeling is cheap, the bottleneck shifts. The phase-breakdown sample reported in Section 7 (Table 4, 38 matched configurations across com-youtube, web-Stanford, and web-it-2004) measures SPIN*'s peel time directly: the median peel cost is 46 ms and the rest of wall-clock — almost entirely CPI construction plus graph load — accounts for a median 99.7% of total time, with no measured cell below 95.4%. The split is most extreme on the largest graph: on web-it-2004 at $s = 3$, SPIN* peels in 54 ms but the full pipeline takes 147 seconds. On web-Stanford at $s = 60$, peel is 2.3 ms out of 7.1 seconds. The pattern is not noise from a few outliers: across the sweep, peel is consistently a sub-percent fraction of wall-clock once SPIN* replaces the iterated full scans of CND. This makes CPI construction the load-bearing cost of the static-state pipeline, and further pipeline-level speedups must therefore parallelise the builder rather than chase peeling tricks.

Why the seed loop can be parallel. The CPI builder follows the standard degeneracy-ordered Bron-Kerbosch search with Tomita pivoting [2, 4, 9, 13]. Each s -clique has one owner: the smallest

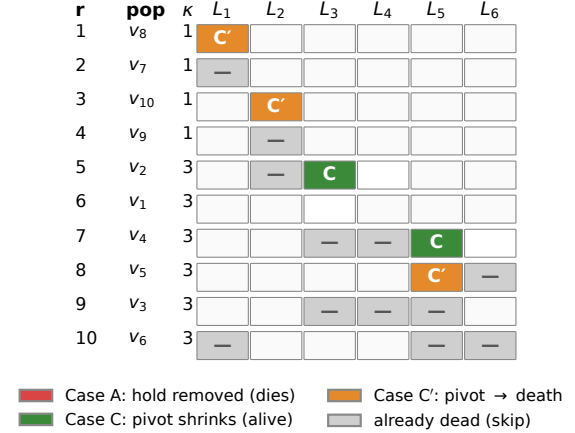


Figure 2: Incremental peel trace on the running example: each row is one pop, each column is one CPI leaf, and each cell is the case under Theorem 3.1 for that (round, leaf). A hold peel kills its leaf (Case A); a pivot peel either shrinks the counter (Case C, alive) or kills the leaf if the counter would fall below η (Case C'). The Round-5 walk-through is in Example 3.1.

vertex of the clique in the degeneracy order. Therefore, the outer loop over seed vertices partitions the emitted s -clique families. No two seeds own the same s -clique.

Static CPI also changes what construction must produce. The mutable baseline builds residual tree state because later peeling may edit that state. Our $r = 1$ pipeline never edits the stored hold or pivot sets. The builder therefore only has to emit a path list and vertex-path incidence arrays. This output can be accumulated in thread-local arenas and merged after all seeds finish.

Thread-local construction. ParaBuildCPI takes G , s , and a thread count. Each worker owns a contiguous seed range, a local path arena, and a generation-stamped membership bitmap. This ownership is enough because seed ranges partition the emitted clique families. While a worker processes its seeds, it writes only to its own arena. After all workers finish, a deterministic merge concatenates the arenas and builds the shared incidence CSR. The expensive work is still the Bron-Kerbosch search under the seeds; the merge is linear in the emitted incidences.

BkEMIT is the same pivot-aware enumeration used by the CPI construction lineage [6, 9]. The difference is the output contract. It streams each emitted path into A_i instead of storing a mutable recursion tree for later residual updates.

Correctness. ParaBuildCPI is a parallel schedule of the same seed-owned enumeration. Seed ownership gives disjoint emitted s -clique families. The deterministic merge changes only the storage order, not the represented path families. The resulting path set satisfies Definition 2.6. Therefore, the support formula of Lemma 2.1 applies unchanged.

Scope. The design parallelizes construction because construction is the phase exposed by the static-CPI pipeline. It does not claim a lower bound showing that SPIN* cannot be parallelized. It also does not claim ideal scaling. The scaling evidence is reported in Table 6.

Algorithm 4: ParaBuildCPI — parallel CPI construction via per-thread path buffers.

Input: Graph G in CSR; clique size s ; thread count T .
Output: Static CPI $\mathcal{P}$ in shared CSR.

```

1  compute degeneracy ordering  $\sigma$  on  $G$ ;
2  launch  $T$  threads and assign each thread  $t$  a contiguous chunk  $\sigma_t$  of seed
   vertices;
3  foreach thread  $t \in \{1, \dots, T\}$  in parallel do
4     $A_t \leftarrow \emptyset$ ; // thread-local path arena
5     $b_t \leftarrow$  generation-stamped bitmap over  $V$ ;
6    foreach seed vertex  $v \in \sigma_t$  do
7      increment generation of  $b_t$ ;
8       $N^+ \leftarrow \{u \in N_G(v) \mid \sigma(u) > \sigma(v)\}$ ;
9       $\text{BrEMIT}(v, N^+, A_t, b_t)$ ; // pivoted BK on  $G[N^+]$ , with  $v$ 
       as a hold
10  $\mathcal{P} \leftarrow$  deterministic concat-and-prefix-sum merge of  $\{A_t\}_{t=1}^T$ ;
11 build incidence CSR over  $\mathcal{P}$  in parallel;
12 return  $\mathcal{P}$ 

```

6 Core-Value Hierarchy Construction

After SPIN^{*}, the pipeline has more than core values. The static CPI survives peeling unchanged, and the deletion log gives the vertices in nondecreasing core-value order. The remaining question is whether these two artifacts together expose the s -clique-connected nucleus hierarchy of Definition 2.4 without rebuilding the index or repeating any clique enumeration. The answer is yes, in a single descending pass over CPI leaves with a union-find structure.

The core numbers define a nested family of vertex sets. For any level k , the super-level set is $V_k = \{v \mid \kappa_s(v) \geq k\}$. As k decreases, vertices enter the set. Definition 2.4 measures connectivity inside V_k through k -admissible s -cliques: two vertices are connected only if they sit in a common surviving s -clique witness. BuildHier constructs the elder-rule join tree of this clique-witnessed filtration directly from CPI leaves.

Leaves as connectors. A CPI leaf P encodes the family of s -cliques $V_h(P) \cup X$ for every η_P -subset $X \subseteq V_p(P)$ (Definition 2.6). Define the *birth level* of P as

$$k_P = \min_{v \in V_h(P) \cup V_p(P)} \kappa_s(v).$$

At level $k \leq k_P$ every leaf vertex has core value $\geq k$, so every encoded s -clique is k -admissible. Any two vertices of the leaf either lie inside one such clique (when $\eta_P \geq 2$) or are linked by a length-two path through a vertex of $V_h(P)$ (when $\eta_P = 1$ and $V_h(P)$ is non-empty); for $V_h(P) = \emptyset$ and $\eta_P \geq 2$, any pair appears together in some s -subset. In every case the leaf vertices form a single s -connected component the moment P becomes admissible. This lets BuildHier treat each leaf as a single hyperedge in a union-find structure: the leaf node is unified with every vertex it contains. Two vertices then end up in the same component if and only if they are s -connected through admissible CPI cliques at the current level.

Reverse scan over leaves. BuildHier processes leaves in nonincreasing k_P , as the level descends through the filtration. A union-find structure spans both vertices and leaf identifiers; processing leaf P at its birth level fires unions $\text{UNION}(P, v)$ for every $v \in V_h(P) \cup V_p(P)$. A birth level is recorded for each component root, so a merge applies the elder rule. Vertices not appearing in any leaf with $k_P \geq 1$ have $\kappa_s(v) = 0$ and are omitted from the hierarchy.

Correctness. The descending scan visits each leaf at exactly the level it first contributes admissible s -cliques. At that level, every

Algorithm 5: BuildHier — elder-rule join tree atop SPIN^{*}.

Input: Graph G , clique size s , CPI $\mathcal{P}$.
Output: Elder-rule join tree T of the s -clique-connected super-level filtration.

```

1   $\kappa_s \leftarrow \text{SPIN}^*(G, s, \mathcal{P})$ ; // Algorithm 3
2  foreach leaf  $P \in \mathcal{P}$  do
3     $k_P \leftarrow \min\{\kappa_s(v) \mid v \in V_h(P) \cup V_p(P)\}$ ;
4  sort  $\mathcal{P}$  by nonincreasing  $k_P$  (ties broken by leaf id);
5  initialise union-find  $U$  over  $V \cup \{P \mid P \in \mathcal{P}\}$ ; birth[ $r$ ]  $\leftarrow +\infty$  for every
   root  $r$ ;
6   $T \leftarrow$  empty forest;
7  foreach leaf  $P$  in sorted order do
8     $R \leftarrow \{\text{FIND}(x) \mid x \in V_h(P) \cup V_p(P) \cup \{P\}\}$ ;
9    foreach root  $r \in R$  do
10     if birth[ $r$ ] =  $+\infty$  then
11       emit a hierarchy node  $n_r$  with  $k_{\text{birth}} \leftarrow k_P$ ; birth[ $r$ ]  $\leftarrow k_P$ ;
12   if  $|R| \geq 2$  then
13     sort  $R$  descending by (birth[ $r$ ],  $|r|$ ); let  $r^*$  be the elder; // elder
       rule
14     foreach  $r \in R \setminus \{r^*\}$  do
15       attach  $n_r$  as child of  $n_{r^*}$  at level  $k_P$  in  $T$ ; // younger dies
16   UNION all of  $R$  into a single component;
17 return  $T$ 

```

vertex it contains is admissible, so the leaf witnesses pairwise s -connectivity for all of them through a common element of $V_h(P)$ (or directly within an s -subset when $V_h(P) = \emptyset$). Conversely, every admissible s -clique at level k belongs to some leaf with $k_P \geq k$, and that leaf has already fired before the scan reaches level k ; so every s -clique-connected merge of Definition 2.4 is performed. When two distinct components meet at a leaf, the elder rule preserves the component with earlier birth and attaches the younger one in the join tree [3].

Complexity. Computing every k_P and sorting leaves costs $O(\Sigma + L \log L)$, where $L = |\mathcal{P}|$. Each leaf fires at most $|V_h(P)| + |V_p(P)|$ union-find operations, summing to $O(\Sigma \alpha(n + L))$ over all leaves. Total time $O(\Sigma \log \Sigma)$, dominated by the sort, with no clique re-enumeration: BuildHier reads each CPI leaf at most twice (once to compute k_P , once to fire unions).

Memory. Beyond the union-find of size $n + L$, the routine needs a list of vertices per leaf to fire unions. SPIN^{*}'s vertex-to-leaf inverse CSR (Section 4) supports a single bucket-CSR pass that materialises the per-leaf lists in $O(\Sigma)$ temporary memory; this scratch is freed on return. The static CPI budget of Section 7 therefore holds throughout peeling proper and is only briefly elevated during the hierarchy emit.

Consequence. The clique-coherent core-value hierarchy is not a separate decomposition problem in this pipeline. It is a byproduct of the static CPI surviving peeling: leaves act as hyperedges of an implicit s -clique witness graph, and a single descending pass with union-find produces the join tree. Methods that mutate the CPI during peeling cannot reuse it for hierarchy emit and need a second clique-enumeration pass [11].

7 Experiments

The previous sections prove the static-CPI invariant and derive two ways to compute the same fixed point: the direct SPIN solver and the practical SPIN^{*} realization. The comparison with REF is therefore a cost comparison, while exactness comes from the invariants and fixed-point equivalence proved earlier. They ask whether the static pipeline changes the cost profile enough to matter. **RQ1** measures

Table 2: Datasets used in Section 7. The last column gives the number of matched successful $(graph, s)$ configurations in the benchmark sweep.

Graph	$ V $	$ E $	# (g, s)
com-amazon [14]	334,863	925,872	29
com-dblp [14]	317,080	1,049,866	109
twitter [14]	81,306	1,342,296	29
web-Stanford [14]	281,903	1,992,636	60
com-youtube [14]	1,134,890	2,987,624	16
web-Google [14]	875,713	4,322,051	39
wiki-Talk [14]	2,394,385	4,659,565	24
web-it-2004 [14]	509,338	7,178,413	429
soc-pokec [14]	1,632,803	22,301,964	24
com-orkut [14]	3,072,441	117,185,083	3
Total			762

end-to-end time and peak memory for **Pure**, the practical static-CPI pipeline. **RQ2** separates construction from peeling, because counter-only peeling shifts rather than removes all cost. **RQ3** asks whether the static construction contract gives useful parallelism. **RQ4** measures SPIN as the direct fixed-point baseline. **RQ5** probes dense synthetic graphs where clique counts grow quickly.

Hardware and software. All experiments run on a dual-socket Intel Xeon Gold 6248R machine with 24 cores and 503 GB DDR4 memory. The operating system is Ubuntu 22.04. The code is compiled with Clang 18.1 using `-O3 -march=native`. It is linked against TBB 2021.10, Boost 1.83, and `robin_hood::unordered_flat_map`. Single-threaded measurements use one core. Parallel-construction measurements use the thread counts reported in Table 6. Wall-clock and resident-set measurements are averaged or summarized over repeated runs as stated in each experiment.

Datasets. Table 2 lists the benchmark graphs. The set covers collaboration, e-commerce, social, and web graphs. The largest graph is com-orkut with 117M edges. The benchmark sweep contains 762 matched successful $(graph, s)$ configurations.

Algorithms and methodology. REF (also called CND) is the mutable-CPI implementation of general (r, s) -nucleus decomposition [9], restricted to $r = 1$ for this comparison. It maintains the CPI as mutable residual state: each peel step deletes the popped vertex from every path that references it, rewrites the affected path records, and updates a hash-indexed reverse map so the next pop can locate its incident paths. The dominant per-pop cost is therefore the residual-index rewrite plus the hash-set rebuild that reflects the deletion; this work is intrinsic to the mutable design and is what SPIN* avoids by leaving the CPI unchanged. **Pure** (SPIN*) is the practical static-CPI pipeline. It constructs the static index, stores flat incidence arrays, and computes core values by incremental path-counter updates. SPIN is evaluated separately because it is the direct fixed-point formulation rather than the practical implementation. All three solvers consume the same input edge list, run on the same hardware, and produce the same κ_s on every matched cell (verified by an offline diff); the comparison isolates the cost of CPI mutability against static-state peeling, not differences in input or correctness.

For every matched $(graph, s)$ pair, we run REF and Pure on the same input. We record end-to-end wall-clock time and peak resident-set memory from `getrusage's ru_maxrss`. End-to-end

Table 3: Memory budget per vertex-path incidence at $\Sigma \gg n$. Baseline numbers are from the mutable-CPI implementation [9]. Pure numbers are sizeof-based accounting for the static layout.

Component	Baseline	Pure
BK recursion tree T	8Σ	0
Path-to-vertex storage (V_h, V_p)	0	4Σ
Per-path vertex set H_p	32Σ	—
Reverse vertex-to-path map	48Σ	—
Vertex-to-path incidence	—	4Σ
Support array	$4n$	$4n$
Heap / bucket array	$24n$	$16n + 8\kappa_{\max}$
Dominant total	$\approx 88\Sigma$	$\approx 10\Sigma$

Table 4: Representative per-phase wall-clock time in ms, median over three runs.

Graph	s	Pure			REF		
		load	build	peel	load	build	peel
com-youtube	3	2081	3863	440.1	1812	3982	789.7
com-youtube	5	2028	3730	273.5	1732	3795	581.6
com-youtube	8	1967	2991	86.3	1638	3053	270.2
com-youtube	12	1689	2701	13.0	1640	2649	148.5
com-youtube	16	1640	2458	8.2	1964	2488	131.8
web-Stanford	3	1251	6602	381.6	1250	6548	521.5
web-Stanford	10	975	5840	130.4	1225	5961	210.1
web-Stanford	20	999	5482	28.9	1228	5591	77.8
web-Stanford	40	1224	5087	9.6	1034	5007	49.6
web-Stanford	60	1224	4900	1.4	1224	4907	31.3
web-it-2004	3	3389	124074	351.0	3378	125347	465.9
web-it-2004	30	3372	122558	63.9	3037	122689	173.7
web-it-2004	100	3407	121891	40.1	3377	121372	135.9
web-it-2004	200	3416	113825	26.1	3406	107516	108.5
web-it-2004	400	3409	7850	4.6	3368	7847	62.7

time includes graph loading, degeneracy ordering, CPI construction, peeling, and output emission. The exactness claim comes from Theorem 3.1, Lemma 4.2, and Lemma 4.3.

RQ1: end-to-end time and memory. The full benchmark contains 762 matched successful configurations. Across the ten graphs, Pure has $7.42\times$ geometric-mean speedup over REF, with $39.41\times$ maximum speedup. The peak-RSS ratio has $1.43\times$ geometric mean and $3.25\times$ maximum. Figure 3 and Figure 4 show where these gains occur.

Figure 5 focuses on the three graphs used in the detailed phase study. It contains 505 matched configurations. On those cells, Pure has $6.16\times$ geometric-mean speedup and $39.41\times$ maximum speedup. Peak-RSS ratio has $1.17\times$ geometric mean and $2.21\times$ maximum.

The memory improvement is explained by the incidence-level budget in Table 3. The mutable baseline stores path hash sets and reverse hash maps. Pure stores flat path-to-vertex and vertex-to-path arrays plus path counters. The analytical per-incidence gap is larger than the observed RSS gap because graph adjacency, allocator overhead, and per-vertex structures do not scale with Σ .

RQ2: where does the time go? The static invariant makes peeling cheap, so the next question is whether construction becomes the dominant cost. The phase study uses 17 configurations on com-youtube, web-Stanford, and web-it-2004, with median times over three runs. Table 4 reports representative phase medians. Table 5 reports the corresponding memory medians.

The phase data supports the bottleneck-shift story. Peeling falls to tens or hundreds of milliseconds on many cells. Construction

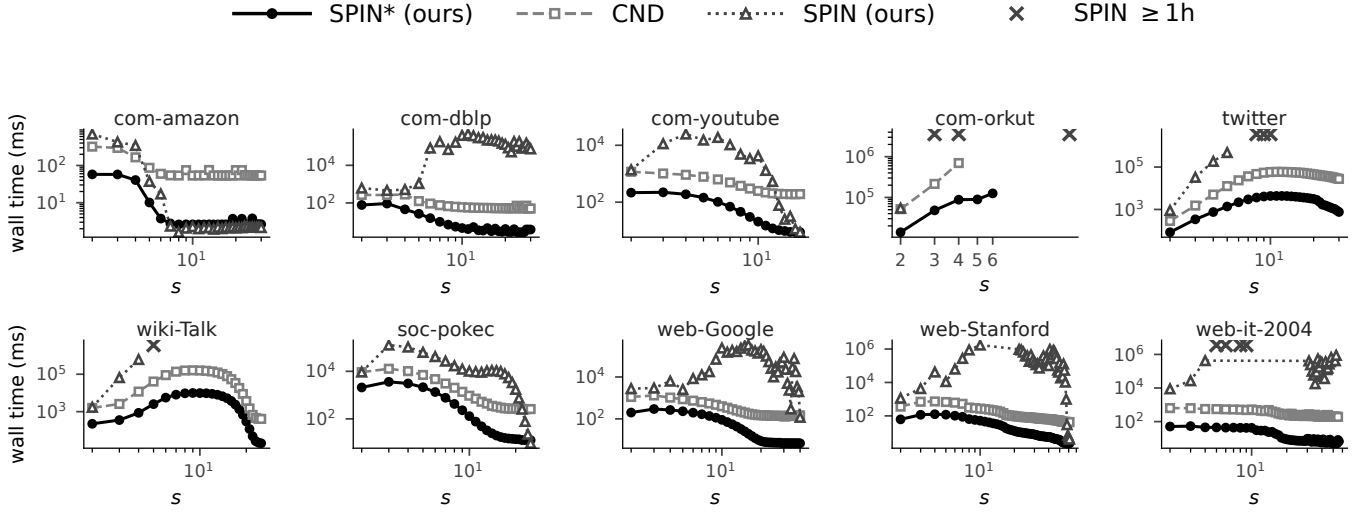


Figure 3: End-to-end wall-clock time as a function of s in the ten-graph comparison. Solid black is SPIN*, the incremental realization; dashed gray is the mutable-CPI reference baseline CND; dotted dark gray is SPIN, the direct iteration. Crosses ($\times$) mark cells where SPIN exceeded the 1h wall-clock cap. SPIN* is consistently the fastest; CND pays residual-index maintenance cost; SPIN pays for repeated full-graph passes and times out on the harder graphs.

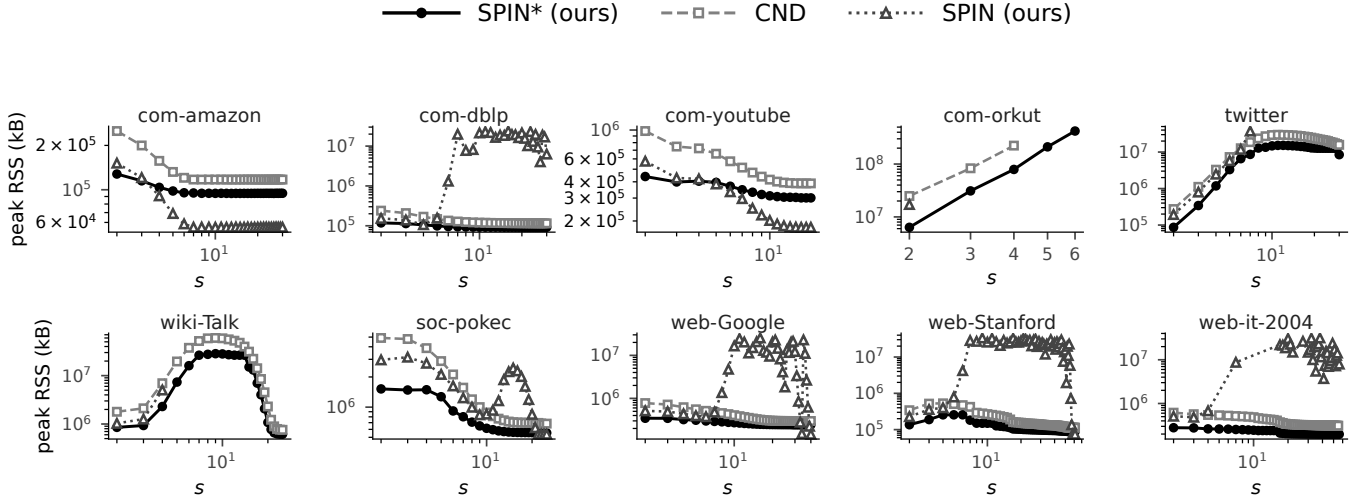


Figure 4: Peak resident-set memory in the ten-graph comparison. Solid black is SPIN*; dashed gray is CND; dotted dark gray is SPIN. The static incidence layout reduces peak RSS when mutable residual path sets dominate memory. SPIN now reads paths from the same flat dual CSR as SPIN*, so its curve tracks SPIN*'s closely instead of carrying the legacy vector-of-vectors overhead.

then becomes a major remaining cost, especially on the web graphs. This is why ParaBuildCPI is part of the algorithmic pipeline rather than a detached engineering detail.

RQ3: does static construction parallelize? ParaBuildCPI exposes seed-level parallelism because each seed can stream its emitted paths into thread-local storage. Table 6 reports build time as the thread count grows from 1 to 24. Each cell is the median of three runs. The table shows strong scaling on large web and social graphs and weaker scaling on smaller overhead-dominated cases.

The correct interpretation is not that construction is perfectly parallel. It is that static output removes hot-path residual-index mutation from the seed walks. This lets the expensive enumeration work scale until memory bandwidth and scheduling overhead become visible.

RQ4: how much does SPIN* optimize over SPIN? SPIN is the literal single-thread solver for the same equations that SPIN* (Pure) realizes by deletion events. The two algorithms compute the same core values (Lemma 4.3), so the comparison isolates the effect of replacing repeated full-graph passes with incremental updates. Both

130 matched cells speedup gmean=14.62× max=44.76× RSS ratio gmean=1.58× max=2.75×

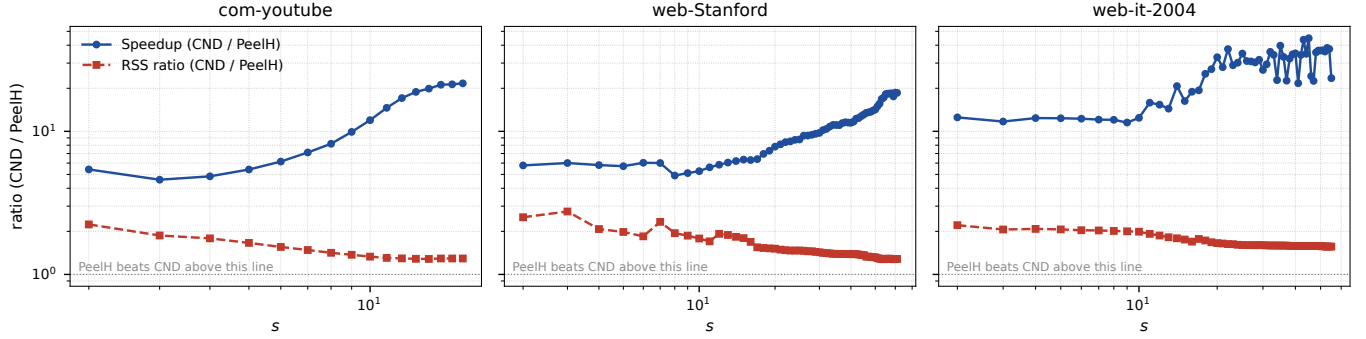


Figure 5: Speedup and peak-RSS ratio on com-youtube, web-Stanford, and web-it-2004. The curves show REF divided by Pure on matched successful configurations. Values above 1 mean Pure is faster or uses less memory.

Table 5: Representative peak resident-set memory in MB, median over three runs.

Graph	s	Pure	REF	Ratio
com-youtube	3	381	729	1.91×
com-youtube	5	315	642	2.04×
com-youtube	8	187	460	2.45×
com-youtube	12	119	385	3.25×
com-youtube	16	114	380	3.33×
web-Stanford	10	176	261	1.48×
web-Stanford	40	70	129	1.85×
web-Stanford	60	53	111	2.09×
web-it-2004	3	425	556	1.31×
web-it-2004	100	207	296	1.43×
web-it-2004	400	139	227	1.63×

Table 6: ParaBuildCPI build time in ms, median over three runs. Speedup is $T = 1$ time divided by $T = 24$ time.

Graph	s	$T = 1$	$T = 2$	$T = 4$	$T = 8$	$T = 16$	$T = 24$	sp.
com-dblp	3	205	131	84	61	56	58	3.5×
com-dblp	15	182	115	73	51	43	45	4.0×
dblp-core30	3	47	36	28	25	22	22	2.1×
com-youtube	3	2644	1516	833	465	275	214	12.4×
com-youtube	16	2418	1367	744	399	224	168	14.4×
web-Stanford	3	5357	2776	1538	834	458	331	16.2×
web-Stanford	60	5196	2697	1483	793	422	297	17.5×
web-it-2004	3	92312	49798	26304	13048	5789	3946	23.4×
web-it-2004	400	83795	45977	24673	10935	5863	3930	21.3×

implementations read paths from the same flat dual CSR; the SPIN reference therefore reflects the algorithmic cost of repeated full-pass fixed-point iteration, not data-structure overhead. Figure 6 pairs SPIN and SPIN* on the matched $(graph, s)$ cells. The geometric-mean peel-time speedup over the full set is two orders of magnitude; the hardest 20 cells (those shown) reach gmean above 96,169× and a maximum near 180,000× on web-Stanford at $s = 44$. On the easiest cells SPIN* can be slightly slower because the bucket-queue setup and live-path bookkeeping add fixed overhead that SPIN’s tight inner loop avoids when one or two passes already converge. Across cells where SPIN hits the 1h cap (notably com-orkut, twitter, wiki-Talk, and web-it-2004), SPIN* finishes the same input in seconds to minutes.

SPIN’s memory profile, with the dual-CSR data structure shared with SPIN*, is essentially identical to SPIN*’s on every paired cell:

Table 7: Representative SPIN runs. Pure time is end-to-end wall-clock time on the same $(graph, s)$ cell.

Graph	s	SPIN	SPIN (s)	Pure (ms)	ratio	RSS (GB)
com-amazon	8	OK	0.9	2.4	367×	0.1
com-dblp	10	OK	391.2	9.8	39782×	20.4
com-youtube	5	OK	21.0	296.1	71×	0.4
soc-pokec	5	OK	99.2	4309.5	23×	2.0
wiki-Talk	5	TIMEOUT	≥ 1800	8645.5	–	–
web-Google	8	OK	23.2	266.0	87×	1.3
web-Stanford	10	OK	1726.8	132.5	13032×	24.5

the geometric-mean RSS ratio is within a few percent, and SPIN is consistently no larger than SPIN*. The reason is structural: SPIN*’s extra state beyond the shared CPI consists of a per-path counter cache (~ 13 bytes per leaf) and a peel-order bucket queue (~ 17 bytes per vertex), which SPIN does not maintain because it scans every vertex each round. The sub-percent memory advantage of SPIN is therefore the price SPIN* pays to avoid repeated full-graph passes; the trade is the textbook “space for time” choice, with the time term dominating by orders of magnitude.

For reference, Table 7 lists representative single-threaded SPIN runs alongside the corresponding SPIN* times.

The result is not a failure of the fixed-point view. It explains why the paper needs both algorithms. SPIN gives the direct semantic formulation. Pure computes the same fixed point by processing only the vertices whose witness counters changed.

RQ5: dense synthetic stress. The benchmark graphs are sparse. The stress test follows the dense synthetic protocol used in prior nucleus-decomposition studies [9, 10]. It generates 1000-vertex graphs by incremental clique injection and sweeps 12 densities from 0.01 to 0.60 for $s \in \{5, 7, 9, 11\}$. Both methods grow quickly as density and s increase. Where both implementations finish, Pure remains below REF in time and memory.

Takeaway. The experiments support the paper’s dependency chain. Pure is faster and uses less peak memory than REF on the 762 matched benchmark cells. The phase breakdown shows that construction becomes a major remaining cost after peeling becomes counter-only. ParaBuildCPI addresses that shifted bottleneck by parallelizing static construction. SPIN confirms why the direct fixed-point view is useful for semantics but unsuitable as the practical solver.

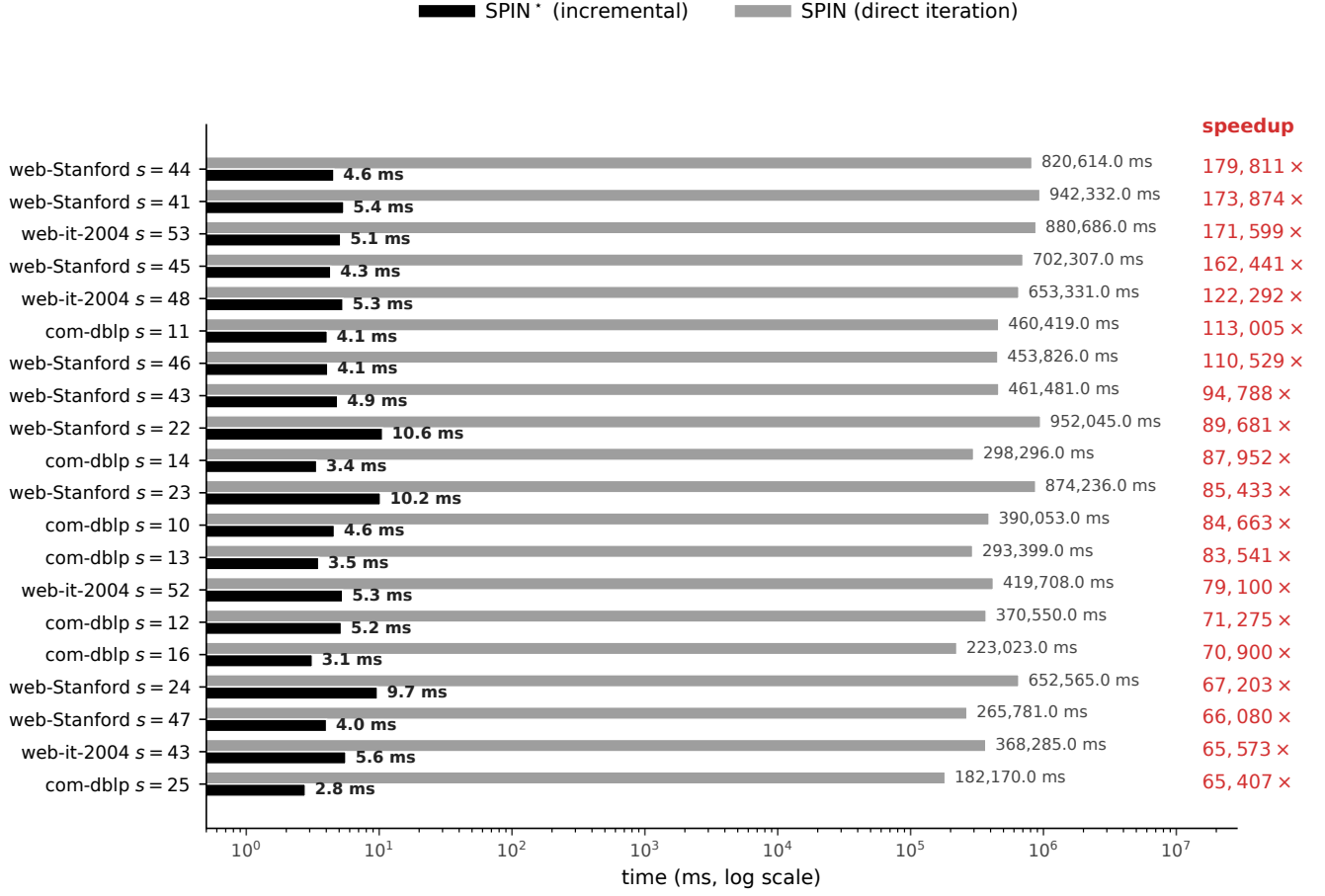


Figure 6: Per-cell wall-clock time of SPIN* (incremental) versus SPIN (direct iteration), top 20 of the matched $(graph, s)$ cells sorted by speedup. Both algorithms compute the same $(1, s)$ -core values; SPIN* avoids redundant full-graph passes by only revisiting vertices whose path counters changed.

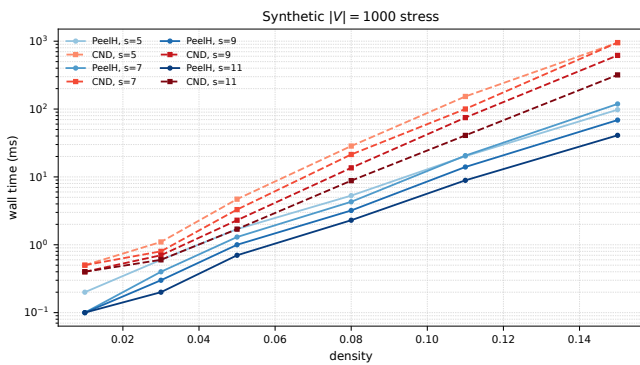


Figure 7: Runtime stress test on synthetic clique-injection graphs with $|V| = 1000$. Solid blue is Pure, and dashed red is REF. Runtime grows rapidly as density and s increase.

8 Case Studies

The experiments measure whether the static pipeline changes cost and scalability. They do not by themselves explain whether the

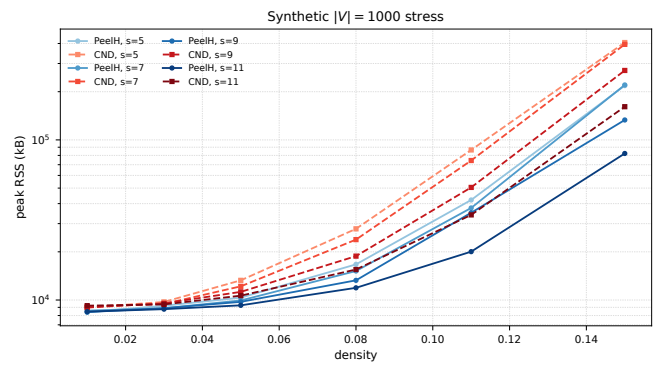


Figure 8: Peak resident-set memory under the same stress-test protocol as Figure 7. The memory gap shows when mutable residual state amplifies dense-clique growth.

specialized $r = 1$ case is worth optimizing. This section therefore asks a practical question: when an analyst wants vertex-level cohesive structure, is $(1, s)$ enough? On the ground-truth tasks reported here, $(1, s)$ matches the quality of higher- r nuclei on the evaluated

Table 8: Active-set size and top- K thresholds on com-dblp as s grows.

s	nonzero $ V $	% of n	max core	top-100 min	top-1k min
2	317,080	100.00	1.13×10^2	1.13×10^2	3.10×10^1
3	269,181	84.89	6.33×10^3	6.33×10^3	4.65×10^2
4	194,957	61.49	2.34×10^5	2.34×10^5	4.50×10^3
5	127,805	40.31	6.44×10^6	6.44×10^6	3.15×10^4
8	36,412	11.48	3.86×10^{10}	3.86×10^{10}	2.63×10^6
10	19,354	6.10	5.97×10^{12}	5.97×10^{12}	2.02×10^7
15	6,767	2.13	2.74×10^{17}	2.74×10^{17}	2.65×10^8
20	3,396	1.07	1.69×10^{21}	1.69×10^{21}	1.41×10^8
25	2,069	0.65	2.18×10^{24}	2.18×10^{24}	2.63×10^6
30	1,358	0.43	7.61×10^{26}	7.61×10^{26}	4.65×10^2

Table 9: Ground-truth quality of $(1, s)$ -cores versus k -core on com-dblp.

Method	top- K	Precision	rec50	tri/v
k -core	10,000	0.504	94	112.3
$(1, 3)$ -core	10,000	0.523	114	113.4
$(1, 5)$ -core	10,000	0.523	114	113.4
$(1, 10)$ -core	10,000	0.523	114	113.4
$(1, 15)$ -core	6,767	0.548	65	154
$(1, 30)$ -core	1,358	0.507	4	523

metrics while costing much less. The claim is empirical and scoped to these datasets and metrics.

Setup. For cross- r comparisons, we use the standard (r, s) -nucleus parameterization [11]. The paper’s algorithmic results apply only to $r = 1$. The $r = 2$ and $r = 3$ results are baselines from the same nucleus family. Because those baselines score edges or triangles, we project their scores to vertices by assigning each vertex the maximum score of an incident r -clique. Vertices are ranked by projected score in descending order, with ties broken uniformly.

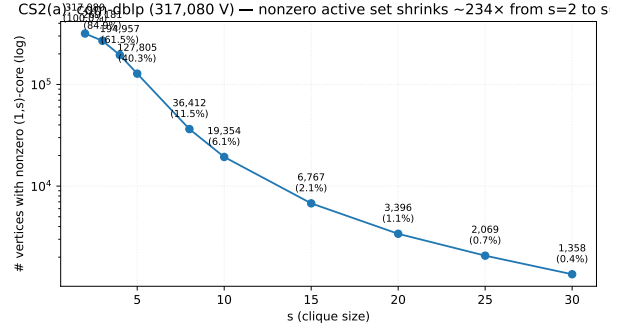
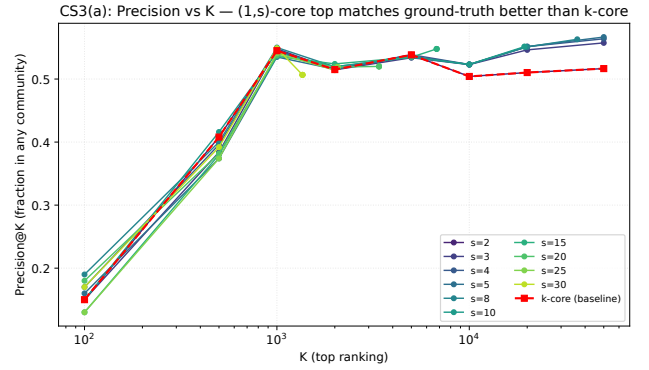
The ground-truth datasets are com-dblp and com-youtube from SNAP [14]. com-dblp has 317,080 vertices, 1,049,866 edges, and 5,000 venue-based ground-truth communities. com-youtube has 1,134,890 vertices, 2,987,624 edges, and 5,000 ground-truth communities.

We use three ranking metrics. Precision@ K is the fraction of top- K vertices that belong to at least one ground-truth community. rec50 is the number of communities with at least 50% of their vertices inside the top- K . Triangle density is the number of triangles per vertex in the induced top- K subgraph.

Case study 1: s is a granularity knob. The first question is what s changes when $r = 1$. On com-dblp, increasing s sharply shrinks the active set while preserving the very top of the ranking. Table 8 reports the active-set size and top-prefix thresholds. Figure 9 shows the monotone shrinkage.

The active set shrinks from all vertices at $s = 2$ to 1,358 vertices at $s = 30$. This is a 234 $\times$ reduction. At the same time, the maximum core value spans many orders of magnitude because high- s supports are binomial counts. The practical interpretation is simple. s controls how aggressively the ranking filters the graph.

Case study 2: $(1, s)$ improves the k -core ranking. On com-dblp, the first non-trivial clique support already improves the ground-truth ranking over k -core. Table 9 reports top-10,000 quality. Figure 10 shows precision over K .

**Figure 9: Active-set size on com-dblp as s varies. Observe that larger s sharply shrinks the vertices with nonzero $(1, s)$ -core value.****Figure 10: Precision@ K on com-dblp. Observe that the $(1, s)$ -core rankings for $s \in \{3, 5, 10, 15\}$ stay above the k -core curve in the plotted range.****Table 10: Cross- r comparison at $s = 10$ on com-dblp.**

Method	Precision	rec50	tri/v	Time (ms)
k -core	0.504	94	112.3	2.3
$(1, 10)$ -core	0.523	114	113.4	13.4
$(2, 10)$ -core	0.529	113	71.8	123.1
$(3, 10)$ -core	0.524	114	113.4	2,379.6
(1 vs. 3)	-0.001	0	0.0	-178 $\times$

$(1, 3)$ -core raises precision from 0.504 to 0.523 and rec50 from 94 to 114. For $s \in \{3, 5, 10\}$, the top-10,000 quality is identical in this table. For larger s , the active set drops below 10,000, so the comparison changes from ranking quality at fixed K to density of the surviving subgraph. $(1, 30)$ -core reaches 523 triangles per vertex on 1,358 vertices.

Case study 3: higher r does not buy quality on DBLP. The next question is whether edge-level or triangle-level nuclei improve the vertex ranking enough to justify their cost. Table 10 fixes $s = 10$ on com-dblp. The $(1, 10)$ -core and $(3, 10)$ -core match on rec50 and triangle density, while $(1, 10)$ is 178 $\times$ faster in the reported run.

The broader grid in Figure 11 shows the same pattern on the reported DBLP settings. The $r = 1$ frontier dominates the $r \in \{2, 3\}$ points in the plotted quality-runtime plane. This does not prove higher r is never useful. It shows that on this vertex-ranking task, increasing r mostly adds cost.

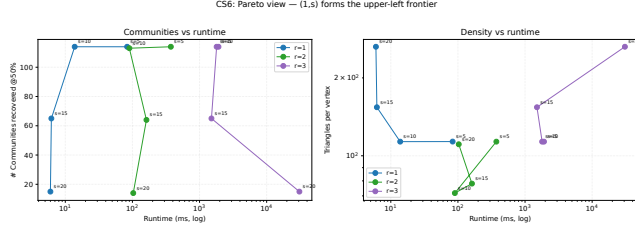


Figure 11: Quality-runtime Pareto plot on com-dblp for $r \in \{1, 2, 3\}$ and $s \in \{5, 10, 15, 20\}$. Observe that the $r = 1$ points lie on the favorable quality-runtime frontier in the DBLP settings.

Table 11: Cross- r comparison on com-youtube.

Method	rec50	tri/v	Time (ms)	vs. $(1, s)$
$(1, 5)$	283	173	274	—
$(2, 5)$	286	161	2,580	9.4×
$(3, 5)$	288	159	13,112	47.9×
$(1, 10)$	49	293	31	—
$(2, 10)$	49	293	319	10.3×
$(3, 10)$	49	293	2,093	67.5×
$(1, 15)$	0 [†]	173	9	—
$(2, 15)$	0 [†]	173	55	6.2×
$(3, 15)$	0 [†]	173	87	9.8×

[†] At $s = 15$, the active set has 95 vertices, below the smallest SNAP community size, so rec50 is zero for every method.

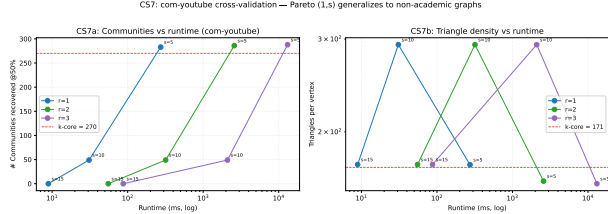


Figure 12: Quality-runtime Pareto plot on com-youtube for $r \in \{1, 2, 3\}$ and $s \in \{5, 10, 15\}$. Observe that $r = 1$ reaches the same quality contours as higher- r settings with lower runtime.

Case study 4: YouTube repeats the pattern. The cross- r comparison is not limited to collaboration data. Table 11 reports the same comparison on com-youtube. At $s = 10$, all three r -values match on rec50 and triangle density, while $r = 1$ is 67.5× faster than $r = 3$.

Figure 12 visualizes the same quality-runtime tradeoff. The $r = 1$ points sit on the same quality contour as higher- r points but with lower runtime.

Case study 5: s changes local resolution. The previous studies use global rankings. The ego-network visualization asks whether s also changes local interpretability. Figure 13 shows the two-hop ego-network of com-dblp vertex 52213 under $s \in \{3, 5, 10\}$. Table 12 reports the component statistics.

At $s = 3$, the ego-network remains one large component. At $s = 10$, it becomes 26 smaller components. This behavior makes s a local resolution knob.

Case study 6: hierarchy profiles differ by domain. BuildHier turns the core values into a $(1, s)$ -clique-connected nucleus hierarchy (Definition 2.4). This makes it possible to compare how quickly

Table 12: Ego-network of com-dblp anchor 52213 under three s values.

s	$ V $	CCs	largest CC	intra-frac	avg cond
3	462	1	462	0.992	0.004
5	335	5	329	0.796	0.121
10	66	26	32	0.340	0.636

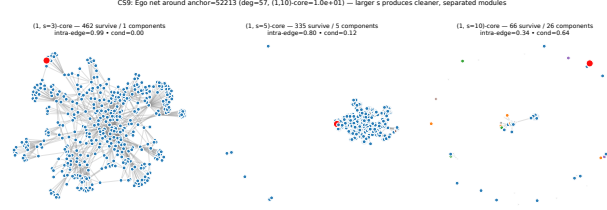


Figure 13: Two-hop ego-network of a com-dblp anchor under $(1, s)$ -decomposition. Vertices are colored by connected component among surviving vertices. Observe that increasing s splits one large local structure into smaller components.

Table 13: Nucleus hierarchy summary. The table reports branches, internal absorption events, and maximum tree depth.

	$s = 3$			$s = 5$			$s = 10$			$s = 15$		
graph	br.	int.	d	br.	int.	d	br.	int.	d	br.	int.	d
com-dblp	10,900	426	4	6,050	278	4	751	20	4	197	8	4
com-youtube	13,507	353	4	714	4	3	12	1	2	1	0	1
web-Stanford	3,455	181	5	1,093	20	3	228	7	3	90	2	2

$(1, s)$ -nucleus join tree (elder-rule) across three domains, $s = 5$ — top-80 branches by persistence

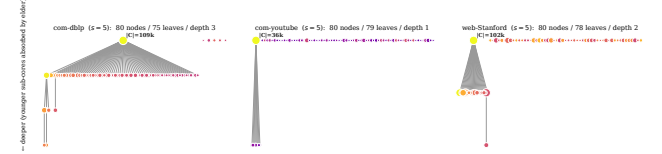


Figure 14: $(1, s)$ -clique-connected nucleus join tree at $s = 5$ across three domains. Observe that the hierarchy shapes differ across collaboration, social, and web graphs.

the nucleus hierarchy collapses across domains. Table 13 reports the three hierarchy statistics used here. Figures 14 and 15 visualize the resulting trees and persistence diagrams.

com-youtube collapses quickly: from 13,507 branches at $s = 3$ to a single branch at $s = 15$. com-dblp keeps the largest branch counts and a constant maximum depth across s , reflecting strong nested clique structure. web-Stanford retains nontrivial branches at every clique size, including $s = 15$. The hierarchy is therefore not only an output format. It gives a domain-level fingerprint of clique-witness nesting.

Takeaway. These studies support the paper's focus on $(1, s)$. The parameter s changes granularity. On the reported DBLP and YouTube ground-truth tasks, $r = 1$ matches higher- r quality metrics while running much faster. The hierarchy and ego-network studies show that the same vertex-centric decomposition also gives interpretable structure beyond a global ranking. The evidence is not a universal claim about all graph mining tasks. It is a practical justification for optimizing the exact $r = 1$ case.

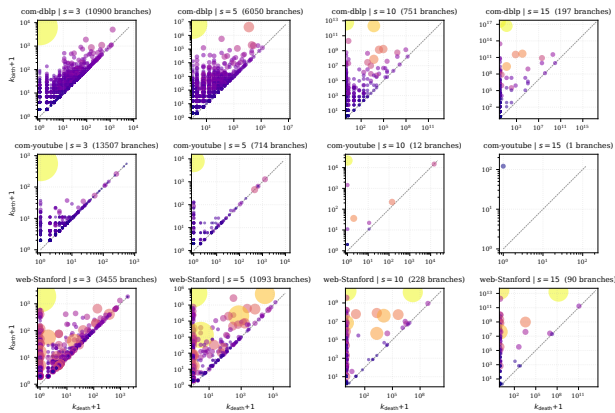
Persistence diagrams of $(1, s)$ -nucleus branches across three domains (rows: graph; cols: s). Marker area $\propto \sqrt{|C|}$ at death; colour = $\log|C|$ at death.

Figure 15: Persistence diagrams across three domains and four s values. Observe that branch persistence changes with both domain and clique size.

9 Related Work

The preceding sections specialize a known compact-clique framework to a narrower state model. The related work is best read through that distinction. Prior work explains how to peel cores and nuclei, how to compactly represent cliques, how to update core values locally or in parallel, and how to build hierarchy outputs. This paper uses those ideas, but its main difference is the static-CPI invariant for $r = 1$.

Enumeration-based core and nucleus decomposition. The k -core was introduced by Seidman [12]. Batagelj and Zaveršnik [1] gave the linear-time peeling algorithm with a bucket queue. Nucleus decomposition generalizes this idea from edges to higher-order clique witnesses [11]. In the general (r, s) setting, peeling removes r -cliques according to their support in s -cliques. That generality is broader than the problem solved here. Our work specializes to the vertex-centric $r = 1$ case and exploits a state invariant that does not hold for general r .

Compact clique representations. Bron-Kerbosch search [2], Tomita pivoting [13], and degeneracy ordering [4] are standard tools for clique enumeration. Pivoter introduced the succinct clique tree as a compact representation for clique counting [6]. The Clique Path Index builds on this line to support general nucleus decomposition without explicit s -clique enumeration, including mutating hold/pivot update rules during peeling [9]. Those representations answer how to compactly encode many cliques. This paper asks a narrower question: for vertex peeling, does the implementation need to materialize those path edits? The static-CPI counter form answers no.

Local and parallel peeling. h -index characterizations of core values were studied for local core computation [5, 7, 8]. SPIN adapts that fixed-point view to the implicit s -clique witnesses encoded by the CPI. SPIN* is the peel-order optimization of the same equations. The difference from prior local h -index work is not the operator itself. The difference is that the witness multiset is represented by static CPI paths and updated by binomial counter deltas.

Parallel construction and clique counting. Parallel clique search usually tries to distribute the expensive recursive enumeration work.

ParaBuildCPI follows the same seed-level decomposition principle. Its specific role in this paper is tied to the static consumer layout. Because the downstream peel never mutates path sets, construction can stream paths into thread-local arenas and merge flat incidence arrays. This is an engineering consequence of the $r = 1$ state model rather than a new clique-enumeration paradigm.

Hierarchical cohesive subgraphs. Nucleus decompositions naturally form hierarchies across core levels [11]. Elder-rule join-tree extraction is standard in computational topology [3]. Prior nucleus pipelines can extract such hierarchies after sorting decomposition output. BuildHier avoids the extra sort because SPIN* already emits a nondecreasing deletion log. The hierarchy result is therefore a byproduct of the chosen peel realization.

Empirical graph mining methodology. The evaluation follows common practice in cohesive-subgraph experiments: SNAP community ground truth, large public graphs, runtime and memory comparisons, and dense synthetic stress tests [9, 10, 14]. The case studies use those benchmarks to test whether the $r = 1$ specialization remains useful as a vertex-ranking tool.

10 Conclusion

The paper began with a tension in general CPI-based nucleus decomposition. The index avoids explicit clique enumeration, but the general algorithm keeps it as mutable residual state. For $(1, s)$ -nucleus decomposition, that mutability is unnecessary. CPI paths do not need to be mutable residual state. Vertex peeling removes vertices but does not delete edges, so the path structure built at the beginning remains valid throughout the peel. A hold removal kills a path contribution. A pivot removal decreases an effective pivot counter. This changes exact decomposition from residual-index maintenance to counter maintenance over a static symbolic index.

The algorithmic consequences follow from that state change. SPIN states the fixed-point computation over the static CPI. SPIN* realizes the same computation in deletion order. ParaBuildCPI addresses construction after cheap peeling exposes it as the next bottleneck. BuildHier reads the sorted deletion log in reverse to recover the hierarchy.

The experiments and case studies support the practical value of the specialization. The reported matched configurations agree with the mutable baseline, and the measured pipeline reduces runtime and memory on the evaluated benchmarks. The case studies show that vertex-centric $(1, s)$ can match higher- r quality metrics on the reported community tasks at much lower cost. The boundary is explicit: the static-CPI invariant is for $r = 1$, not for general (r, s) .

A natural next step is to identify restricted $r \geq 2$ settings where enough path structure survives to support counter-only updates.

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